Multi-Scale Coupling Between LFP and EMG in Mice by Low-Intensity Pulsed Ultrasound Stimulation With Different Number of Tone-Burst

Ziqiang Jin, Xiaoling Chen, Zechuan Du, Yi Yuan, Xiaoli Li, and Ping Xie

***Abstract*— Low-intensity pulsed ultrasound stimulation (LIPUS) as a non-invasive, high-spatial resolution and high penetration depth brain modulation technology has been used for modulating neuromuscular function. However, the modulation of neural electrical signal changes in the neuromuscular system by LIPUS remains to be explored. In this study, we stimulated the mouse brain motor cortex by LIPUS with different number of tone burst (NTB) and recorded the local field potential (LFP) signals of the target region and electromyography (EMG) of tail muscle. Multi-Scale Transfer Entropy (MSTE) analysis method was used to explore the multi-scale synchronization characteristics and functional cortico-muscular coupling(FCMC) strength changes of mice LFP-EMG before and after LIPUS under different NTBs. The results show that the MSTE of LFP-EMG before and after LIPUS stimulation was higher than that of EMG-LFP. After adding multi-scale, MSTE has a significant relationship with time scales. When NTB = 200, the scale of extremum is the largest. There was a fitting intersection between LFP-EMG and EMG-LFP scale 7-21 before and after stimulation. After scale averaging, the LFP-EMG after stimulation was lower than that before stimulation, and the EMG-LFP after stimulation was higher than that before stimulation. Conclusion: There is a significant correlation between NTB and time scale before and after LIPUS, as well as upward and downward. Consequently, This study used FCMC methods to study different NTBs and multi-scale relationships, provides new variables from LIPUS parameters and analysis, and provides new reference for clinical applications of LIPUS.**

***Index Terms*— Low-intensity pulsed ultrasound stimulation, functional cortico-muscular coupling, local field potential, electromyography, multi-scale transfer entropy.**

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Ziqiang Jin and Zechuan Du are with the Institute of Electric Engineering, Yanshan University, Qinhuangdao, Hebei 066004, China (e-mail: ziqiangjin@ysu.edu.cn; duzechuan@stumail.ysu.edu.cn).

Xiaoling Chen, Yi Yuan, and Ping Xie are with the Key Laboratory of Intelligent Rehabilitation and Neuromodulation of Hebei Province, Institute of Electrical Engineering, Yanshan University, Qinhuangdao, Hebei 066004, China (e-mail: xlchen@ysu.edu.cn; yuanyi513@163.com; pingx@ysu.edu.cn).

Xiaoli Li is with the National Key Laboratory of Cognitive Neuroscience and Learning, Beijing Normal University, Beijing 100875, China (e-mail: xiaoli@bnu.edu.cn).

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I. Introduction

IN RECENT years, low-intensity transcranial ultrasound stimulation (LIPUS) has received increasing attention as an effective neuroregulatory technique [1], [2]. The difference between LIPUS and other neuroregulatory technologies lies in its non-invasive nature, high spatial resolution, and penetrability [3], [4], [5]. Many studies have been conducted to analyze the effects of LIPUS on neuronal activity in the brain by observing cerebral blood flow, brain tissue slices, and electroencephalogram signals [6], [7], [8], [9]. LIPUS induces motor behavior in mice, such as the movement of whiskers and tails, and alters the firing rate of neurons [10]. Legon et al.'s study showed that ultrasound stimulation significantly regulates the amplitude of primary somatosensory cortex evoked potentials through the electroencephalogram(EEG) spectrum, enhancing the human body's ability to recognize sensations [11], [12]. LIPUS changes the power spectrum, nonlinear dynamics, and phase amplitude coupling of local field potential (LFP) with different rhythms; And aging brain hemodynamics and cerebral oxygen metabolism, such as increasing cerebral blood flow, accelerating cerebral oxygen metabolism, and enhancing neurovascular coupling [13], [14]. However, few studies have focused on the coupling between neuronal activity and motion control systems. Therefore, it is necessary to conduct a comprehensive analysis and exploration of the EEG and electromyography (EMG) characteristics under LIPUS.

LIPUS can regulate neural oscillations and cause significant muscle contractions, so it is necessary to study whether there is information interaction between LIPUS induced motor cortex and target muscles. Functional cortical muscle coupling (FCMC)is an effective method for studying the relationship between EEG and EMG in the motion control

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circuit system. Wang et al. demonstrated in the laboratory that the parameters of LIPUS can affect the strength of FCMC from the perspective of time-frequency characteristics and phase synchronization [18]. The above research preliminarily confirms that neuromuscular coupling analysis helps to understand and evaluate the regulatory mechanism and intervention effect of LIPUS on the neuromuscular system [19]. FCMC has both upward and downward relationship, and the motor control commands generated by the cortex act on the effector muscles through neural circuits, which can then provide feedback signals to the motor cortex. Transfer entropy (TE) as an effective method in FCMC analysis, can effectively characterize the interaction of nonlinear features and be used to study the effect of LIPUS on the cortical and muscle conversion intensity of mice with different number of tone bursts (NTB), and the resulting interaction. FCMC is significantly enhanced in different directions of NTB [20], [21]. As NTB is a parameter related to stimulation time, this study also focuses on multi-scale research in the time domain around NTB parameters.

Due to the complex multi-scale characteristics of brain function in both temporal and spatial domains, neural oscillations and neuromuscular coupling in motion control systems also exhibit certain multi-scale characteristics. To this end, Costa proposed the multi-scale entropy theory and applied it to the study of heart rate variability (HRV), finding that multi-scale entropy can effectively represent the two diseases and health status of congestive heart failure (CHF) and atrial fibrillation (AF) [22]. Thurisingham et al. used the same method to study cardiac data and demonstrated that sampling time and time scale have an impact on multi-scale entropy characteristics. Due to the current sensitivity of nonlinear time series analysis algorithms to the selection of sequence length and parameters in the algorithm [23]. Xie et al. proposed a multi-scale entropy transfer method to obtain data of different scales and functions [24]. To explore the characteristics of neuromuscular coupling under LIPUS, we used a multi-scale transfer entropy method to calculate the LFP and EMG before and after ultrasound stimulation across various time scales and NTBs.

In summary, the motivation of this study is to investigate the regulation of LIPUS and use LIPUS with different NTBs to stimulate the tail motor cortex of healthy mice. We synchronously recorded the LFP of the tail motor cortex and the EMG of the tail under different parameters. The multi-scale transfer entropy method is used to analyze the coupling relationship between LFP and EMG, which is conducted at multiple scales. By adding time scale variables, the regulatory mechanism of LIPUS can be better analyzed and understood.

II. Material and Methods

A. Animal Anesthesia and Surgery

Twelve male BALB/c mice, weighing between 20 to 25 grams were utilized for the experimental procedures (Beijing Vital River Laboratory Animal Technology Co., Ltd.). These procedures adhered to the guidelines set forth by the Animal Ethics and Administrative Council at Yanshan University in Hebei Province, People's Republic of China. The mice were administered surgical anesthesia using sodium pentobarbital at a concentration intraperitoneal injection(3%, 5mg/100 g, i.p.). The mice were fixed on a stereotactic device (6800268030, RWD Co., China). The hair covering the scalp was shaven and washed with normal saline. The scalp was cut along the midline of skull, and the exposed tissue and periosteum were cleaned. According to the brain atlas, the motor cortex of mice was located, and a ∼1mm * 1mm window was drilled to expose the cerebral cortex of mice. At the end of the experiment, the mice were killed with an excessive anesthetic.

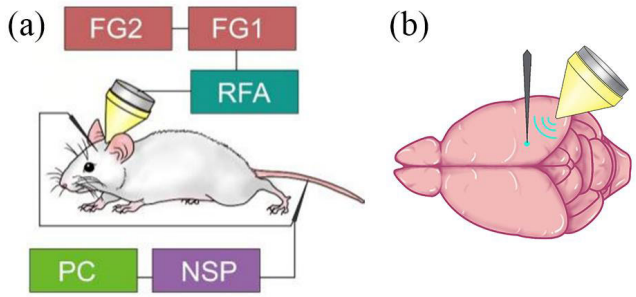

Fig. 1. The experimental system and Stimulus Location (a) The experimental system (b) Stimulus Location (FG:function generator,RFA:RF amplifier,NSP:neural signal processor,PC:personal computer).

B. LIPUS Experimental Setup

As shown in Fig. 1(a), in the LIPUS system, a pair of interconnected arbitrary function generators were deployed to produce pulsed waveforms(AFG3022C, Tektronix, USA). The primary function generator, responsible for generating square wave signals to initiate the second generator, was tasked with regulating the pulse repetition frequency (PRF) and the stimulation duration (SD). Its counterpart, the second function generator, was charged with adjusting the parameters of the ultrasound, namely the fundamental frequency (FF), the duty cycle (DC), NTB, and the spatial peak pulse average intensity (Isppa). The pulsed output from this second generator was then boosted by a linear power amplifier(E&I240L, ENI Inc., USA). In this paper, the parameter setting is Isppa = 1.1 W/cm^2, SD = 3.6s, PRF = 1 kHz, FF = 50%, DC = 250, NTB = 100,150,200,250,300. Subsequently, the raw local field potential (LFP) signals were captured using tungsten microelectrodes(WE50030.1B10, MicroProbe, USA), which were strategically positioned within the region of stimulation as depicted in Fig 1 (b).

C. Data Acquisition

Crude LFPs were obtained through the use of tungsten microelectrodes(WE50030.1B10, MicroProbe, USA), which were precisely placed into the designated stimulation area as illustrated in Fig 1. Concurrently, EMG signals were captured by a single-channel microelectrode(WE50030.1b10 probe, USA) designed for raw EMG signal recording. These microelectrodes, connected to a dual-channel preamplifier, were utilized to record LFP signals. The analog signals from

the preamplifier (63 386, A-M SYSTEMS Inc., USA)were then digitized by a signal processing unit before being sent to a computer system for subsequent data storage and analysis(Cerebus Data Acquisition System, Blackrock Microsystems, USA). A sampling rate of 2 kHz was employed for high-fidelity signal capture. The arrangement ensured a 60° angle between the LIPUS application and the microelectrode placement. Additionally, a collimator, attached to the flat ultrasound transducer, was precisely targeted at the motor cortex located within the tail region of the mice.

D. Data Processing and Statistics

To avoid power frequency interference in original LFPs and EMG records, a notch filter is used to remove 50 Hz power frequency signals, and an adaptive high pass filter is used to remove baseline drift. The data period is 4 s, 0.5-200 Hz band-pass filter is used for LFPs, and 10-200 Hz band-pass filter is used for EMG. The signals before and after LIPUS were intercepted. The data were collected for each group of 30 trials, and the first and the last were manually raised to stack and average the multiple trials.

III. Multiscale Transfer Entropy of Neuromuscular Function Coupling Analysis

A. Multiscale Transfer Entropy

The interaction between LFPs and EMG can reflect the relationship between the motor cortex and effector muscles. To quantitatively study the nonlinear synchronous coupling characteristics of LFPs and EMG on multiple time scales. We choose the multiscale transfer entropy method (MSTE). Firstly, LFPs and EMG are multi-scale to display the information on different time scales. The multi-scale transfer entropy of LFPs and EMG is calculated, and the functional coupling characteristics of LFPs and EMG on different time scales are obtained.

The preprocessed LFPs and EMG are intercepted from the data after LIPUS. Use the coarse-grained method to multi-scale the signal. $X = \{x_1, x_2, \cdots, x_i, \cdots, x_M\}$, $Y = \{y_1, y_2, \cdots, y_i, \cdots, y_M\}$. The formula is as follows:

$$X_j^{(s)} = \sum_{i=(j-1)s+1}^{js} x_i \left(1 \le j \le \frac{M}{S}\right) \tag{1}$$

$$Y_j^{(s)} = \sum_{i=(j-1)s+1}^{js} y_i \left(1 \le j \le \frac{M}{S}\right) \tag{2}$$

In the formula, s is the number of time scales, and j is the coarse-grained signals X and Y. The sequence number of the sequence, M is the length of the sequence X and Y.

Xs and Ys respectively represent LFP and EMG of differentscales s. Based on the definition of transfer entropy, a multi-scale transfer entropy TEs-LFP-EMG from Xs to Ys is constructed. The formula is as follows:

$$TE_{s-LFP-EMG} = \sum_{y_{t+u}/y_t x_t} p(y_{t+u}, y_t, x_t) \times \log \frac{p(y_{t+u}, y_t, x_t)p(y_t)}{p(y_{t+u}, y_t)p(y_t, x_t)} \tag{3}$$

In the formula, t is the discrete-time index; u is the prediction time; x_t, y_t are the delay vectors under the scales; p (·) is the joint probability between the variables. TEs-LFP-EMG means LFP on the s scale The transfer entropy value to EMG. Similarly, the multi-scale transfer entropy TEs-EMG-LFP from Ys to Xs is expressed as:

$$TE_{s-EMG-LFP} = \sum_{x_{t+u}/y_t x_t} p(x_{t+u}, x_t, y_t) \times \log \frac{p(y_{t+u}, y_t, x_t)p(y_t)}{p(y_{t+u}, y_t)p(y_t, x_t)} \tag{4}$$

TEs-LFP-EMG represents the transfer entropy of EMG to LFP in the s scale. For convenience, this paper defines the multi-scale transfer entropy calculated by equations (3) and (4) as MSTEs-LFP-EMG and MSTEs-EMG-LFP.

$$MSTE_{s-LFP-EMG}^{NTB=100} = \{MSTE_{1-LFP-EMG}, MSTE_{2-LFP-EMG}, \cdots, MSTE_{30-LFP-EMG}\} \tag{5}$$

$$MSTE_{s-EMG-LFP}^{NTB=100} = \{MSTE_{1-EMG-LFP}, MSTE_{2-EMG-LFP}, \cdots, MSTE_{30-EMG-LFP}\} \tag{6}$$

In the formula, NTB = 100, which means the NTB is 100. MSTEs-LFP-EMG and MSTEs-LFP-EMG represent the transfer entropy in the coupling directions of LFP-EMG and EMG-LFP respectively.

B. Statistical Analysis

The Kruskal–Wallis test was used to test for statistical significance. Differences were considered significant at p-values < 0.05. Statistical analyses were conducted using MATLAB software.

IV. Results

A. Results for the One Scale

As shown in Fig. 3, the results of MSTE before and after stimulation of LFP-EMG is higher than that of MSTE before and after stimulation of EMG-LFP at five NTBs. Before stimulation, MSTE of LFP-EMG and MSTE of EMG-LFP increased with the increase of NTB. It reaches a peak at NTB = 250. After stimulation, the MSTE of LFP-EMG first increased and then decreased with the increase of NTB. After stimulation, MSTE of EMG-LFP decreased with the increase of NTB.

B. Results for the 30 Scales

As shown in Fig. 4, EMG-LFP before and after stimulation were compared, and the scatter trend was constructed by the fitting curve.

When NTB = 100, the fitting curves of LFP-EMG and EMG-LFP appeared intersection at scale 18. The MSTE of LFP-EMG is higher than that of EMG-LFP before scale 18, and that of EMG-LFP is higher than that of LFP-EMG after scale 18. The MSTE of LFP-EMG was higher than that of EMG-LFP before scale 7, and the MSTE of EMG-LFP after stimulation was higher than that of LFP-EMG after scale 7. The MSTE of LFP-EMG after stimulation was higher than that

before stimulation. The MSTE of EMG-LFP after stimulation was higher than that before stimulation.

When NTB = 150, the fitting curves of LFP-EMG and EMG-LFP before stimulation appeared intersection at scale 20. the MSTE of LFP-EMG was higher than that of EMG-LFP before stimulation before scale 20, and the MSTE of EMG-LFP was higher than that of LFP-EMG before stimulation after scale 20. The curve of LFP-EMG after stimulation and EMG-LFP after stimulation intersected at scale 7. The MSTE of LFP-EMG was higher than that of EMG-LFP after stimulation before scale 7. The MSTE of EMG-LFP was higher than that of LFP-EMG after stimulation after scale 7. The MSTE of LFP-EMG before stimulation was higher than that of LFP-EMG after stimulation. The MSTE of EMG-LFP after stimulation was higher than that of EMG-LFP before stimulation.

When NTB = 200, the fitting curves of LFP-EMG and EMG-LFP before stimulation appeared intersection at scale 10 and scale 22. The MSTE of LFP-EMG was higher than that of EMG-LFP before stimulation before scale 10. The MSTE of LFP-EMG was higher than that of LFP-EMG before stimulation between scale 10 and scale 22. The MSTE of LFP-EMG was higher than that of EMG-LFP before stimulation after scale 22. Although there was a crisscross before and after stimulation, the LFP-EMG before stimulation was higher than that after stimulation. Although there was a crisscross before and after stimulation, the EMG-LFP after stimulation was higher than that before stimulation.

When NTB = 250, the fitting curve of LFP-EMG and EMG-LFP before stimulation appeared intersection at scale 11 and scale 23. The MSTE of LFP-EMG was higher than that of EMG-LFP before stimulation before scale 11. The MSTE of LFP-EMG was higher than that of LFP-EMG before stimulation from scale 11 to scale 23. The MSTE of LFP-EMG was higher than that of EMG-LFP before stimulation after scale 23. The LFP-EMG before stimulation was higher than that after stimulation. The EMG-LFP after stimulation was higher than that before stimulation.

When NTB = 300, The fitting curves of LFP-EMG and EMG-LFP before stimulation appeared intersection at scale 8 and scale 21. the MSTE of LFP-EMG was higher than that of EMG-LFP before stimulation before scale 8. The MSTE of EMG-LFP was higher than that of LFP-EMG before stimulation between scale 8 and scale 21. The MSTE of LFP-EMG was higher than that of EMG-LFP before stimulation after scale 21. The LFP-EMG before stimulation was higher than that after stimulation. The EMG-LFP after stimulation was higher than that before stimulation.

C. Results of Extremum Scale

As shown in Fig. 5, when NTB = 100, the peak scale of LFP-EMG before stimulation is 23, peak scale of EMG-LFP before stimulation is 25, of LFP-EMG after stimulation is 11, of EMG-LFP after stimulation is 16. When NTB = 150, the peak scale of LFP-EMG before stimulation is 17, of EMG-LFP before stimulation is 22, of LFP-EMG after stimulation is 16, of EMG-LFP after stimulation is 20.

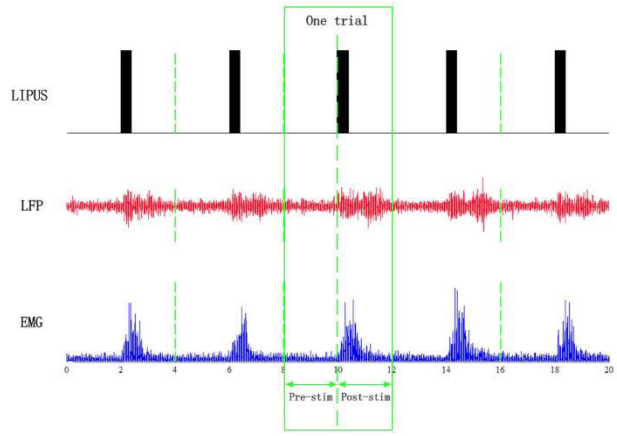

Fig. 2. LIPUS and the LFP and EMG signals after preprocessed.

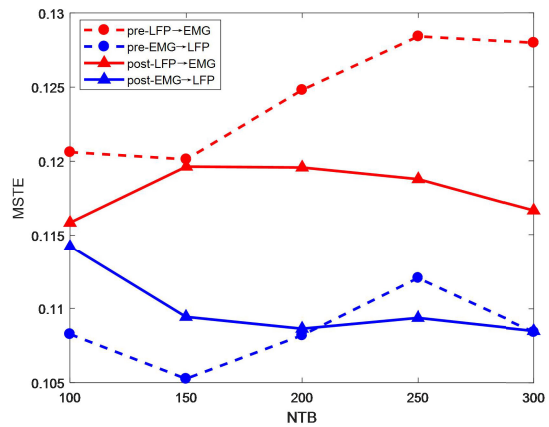

Fig. 3. The results of MSTE(LFP-EMG,EMG-LFP) before and after LIPUS at NTB = 100, 150, 200, 250, 300.

When NTB = 200, the peak scale of LFP-EMG before stimulation is 27, of EMG-LFP before stimulation is 25, of LFP-EMG after stimulation is 19, of EMG-LFP after stimulation is 24. When NTB = 250, the peak scale of LFP-EMG before stimulation is 23, of EMG-LFP before stimulation is 16, of LFP-EMG after stimulation is 18, of EMG-LFP after stimulation is 22. When NTB = 300, the peak scale of LFP-EMG before stimulation is 16, of EMG-LFP before stimulation is 21, of LFP-EMG after stimulation is 25, of EMG-LFP after stimulation is 19. Therefore, the peak scale of LFP-EMG before stimulation and EMG-LFP after stimulation increased and then decreased with the increase of NTB.

D. Results of MSTE With Mean Value of Each Scale

As shown in Fig. 6, the mean value of 30 scales is calculated and the error bar is added. The discreteness of LFP-EMG was higher than that of EMG-LFP before and after stimulation. The mean value of each scale before stimulation was higher in the LFP-EMG than in the EMG-LFP. The mean value of each scale after stimulation was higher in the EMG-LFP than in the LFP-EMG. The results of MSTE with the mean value of each scale NTB = 100 > 250 > 200 > 150 > 300. (mean ± S.D., N = 12, *$p < 0.05$, Kruskal–Wallis test)

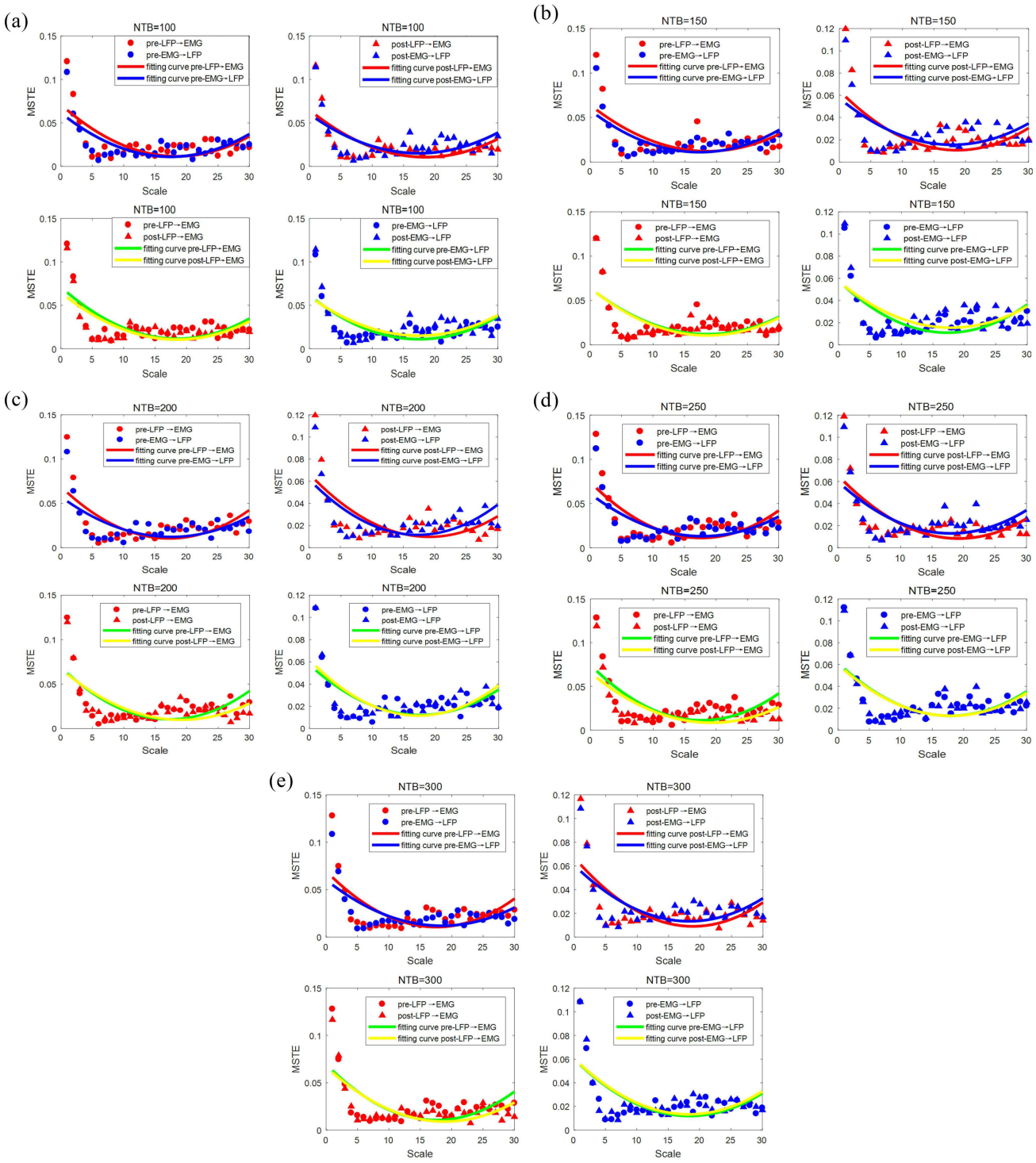

Fig. 4. The results of MSTE (LFP-EMG, EMG-LFP) before and after LIPUS at (a) NTB = 100, (b) NTB = 150, (c) NTB = 200, (d) NTB = 250, (e) NTB = 300.

E. Results of MSTE Before and After LIPUS With Scale Merging

As shown in Fig. 7, the mean value of 30 scales is calculated and the error bar is added. The LFP-EMG before stimulation and the EMG-LFP after stimulation were both higher than before the EMG-LFP before stimulation and the LFP-EMG after stimulation. The stability was NTB =200>150>100>300>250. (mean ± S.D., N = 12, $^{*}p < 0.05$, Kruskal–Wallis test)

F. Results of MSTE With Mean Value of Each Scale

As shown in Fig. 8, the mean MSTE of (LFP-EMG, EMG-LFP) before and after LIPUS with 30 scales at NTB =100, 150, 200, 250, 300. The results are similar to those shown

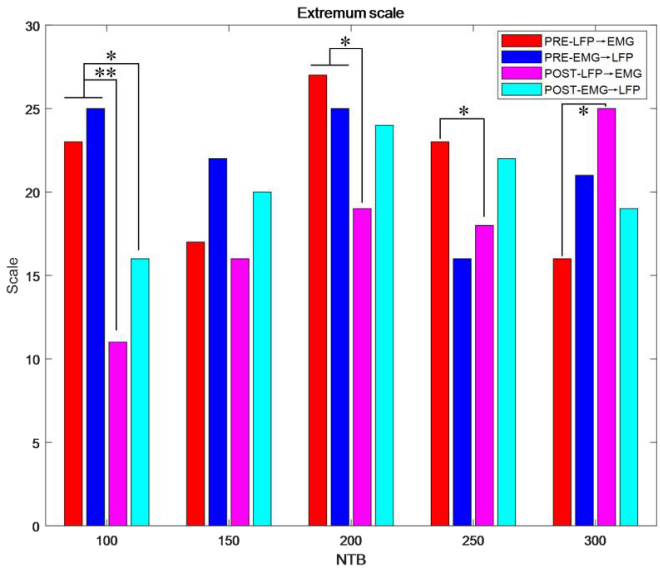

Fig. 5. The results of extremum scale of MSTE (LFP-EMG, EMG-LFP) before and after LIPUS at NTB = 100, 150, 200, 250, 300 (mean ± S.D., N = 12, *p < 0.05, **p<0.01, Kruskal–Wallis test).

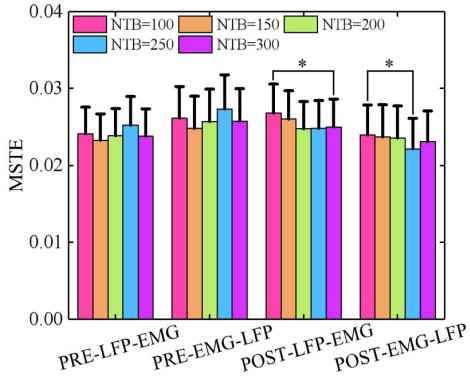

Fig. 6. The results of MSTE(LFP-EMG,EMG-LFP) with mean value of each scale at NTB = 100, 150, 200, 250, 300, before and after stimulation(mean ± S.D., N = 12, *p < 0.05, Kruskal–Wallis test).

in Figure. 2. MSTE with the increase of scale is decreases rapidly, then increases slowly, and tends to be stable,(NTB =100>250>200>150>300).

G. Results of MSTE With Mean Value of Each Scale

As shown in Fig. 9, the mean MSTE of mean MSTE of 5 NTBs with 30 scales (LFP-EMG, EMG-LFP) before and after LIPUS. The results are similar to those shown in Fig. 2. MSTE with the increase of scale is decreases rapidly, then increases slowly, and finally tends to be stable. MSTE was less than the state at the mean of 5 NTBs, indicating that the influence of NTBs on MSTE was higher than that before and after stimulation. MSTE (pre-LFP-EMG>post EMG-LFP>pre-EMG-LFP>post LFP-EMG).

V. Discussion

LIPUS has become an increasingly valuable method that can make up for the deficiencies of existing neuromodulation technologies. It can target and activate brain regions non-invasively. Many studies have used transcranial ultrasound to

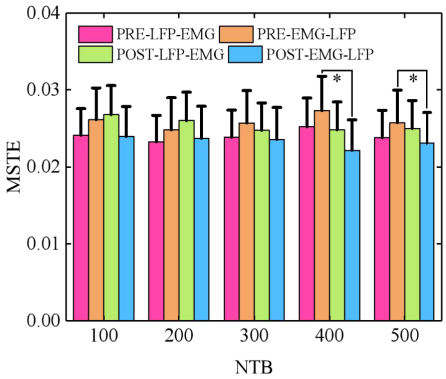

Fig. 7. The results of MSTE (LFP-EMG, EMG-LFP) before and after LIPUS with scale merging at NTB = 100, 150, 200, 250, 300 (mean ± S.D., N = 12, *p < 0.05, Kruskal–Wallis test).

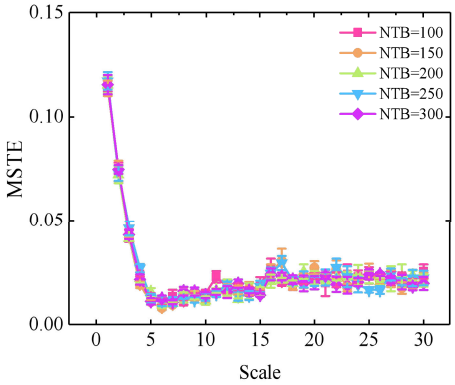

Fig. 8. The results of mean MSTE of (LFP-EMG, EMG-LFP) before and after LIPUS with 30 scales at NTB = 100, 150, 200, 250, 300.

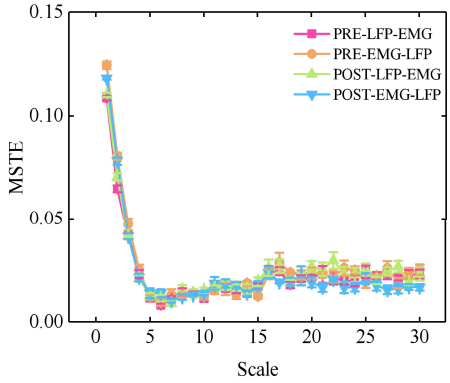

Fig. 9. The results of mean MSTE of 5 NTBs with 30 scales (LFP-EMG, EMG-LFP) before and after LIPUS.

stimulate the primary motor cortex of animals, and significant results have been achieved [25], [26], [27]. EMG collection confirmed that transcranial ultrasound stimulation can induce motor responses. In addition, there is indeed a coupling relationship between cortical neuron oscillation and effector muscle activity [28], [29], [30], [31], [32]. FCMC can be used as an evaluation indicator, so it can be used to evaluate neuromodulation technologies such as transcranial magnetic stimulation, transcranial direct current stimulation and deep brain stimulation. Therefore, we believe that FCMC is also suitable for evaluating LIPUS. In this study, the parameters

of LIPUS have an impact on the LFP and EMG before and after stimulation [33], [34], [35]. NTB is the parameter of the duration of LIPUS stimulation [21]. The greater the NTB, the greater the coupling strength. After adding time scale, MSTE and NTB do not have significant law, but time scale and MSTE have significant relationship. It shows that the time scale can be used as an important index for multi-scale analysis of FCMC in the future [36], [37], [38]. Further comparison of the two directions before and after stimulation, LFP-EMG and EMG-LFP, the fitting scale also gradually increases with the increase of NTB. It can be boldly speculated that the MSTE of NTB to the stimulated emg-lfp is greater than the stimulated lfp-emg. The possible reason is that LIPUS directly stimulates the motor cortex S1 of mice, resulting in tail muscle movement, but not spontaneous motor instructions [14], [25], [39].

When the time scale was 1, LFP-EMG was higher than EMG-LFP before and after stimulation. After the average of the scale, the LFP-EMG after stimulation is lower than that before stimulation, and the principle that the EMG-LFP after stimulation is higher than that before stimulation may be similar.

In this experiment, mice were subjected to mild anesthesia, and anesthesia would inevitably have an impact on the regulatory results of LIPUS [40]. The focus of this article is to analyze the direction of LFP-EMG and EMG-LFP before and after stimulation, as well as the time scale and other issues. In future research, the relationship between anesthesia and LIPUS can be further analyzed.

There is a correlation between MSTE and time scale. With the increase of time scale, MSTE first increases rapidly, and then decreases rapidly, finally tends to be stable. This situation can be attributed to the multi-time scale function itself. When the scale is less than 7, MSTE is greatly affected by the scale and decreases rapidly. When the scale is greater than 7, MSTE tends to be stable. It can provide a limited range for future multi-scale analysis as a reference.

The proposal of multi-scale transfer entropy makes it possible to study the multi-scale characteristics of two coupled signals, and has been applied to the coupling analysis of neuromuscular function. LFP and EMG collected before and after transcranial ultrasound stimulation are multi-scale in time domain, and the time scale is uncertain, which cannot provide additional perspective. Using MSTE to quantify FCMC in future LIPUS will help to further understand the impact of LIPUS on neural rehabilitation and quantify the patient's condition, which will help to adjust the parameters of LIPUS. It will improve the development and evaluation of new treatment interventions.

VI. Conclusion

The study analyzed the neuromodulatory effect of LIPUS was analyzed through the FCMC between the motor cortex and tail muscles of mice, and it was quantified by the method of multiscale transfer entropy. The results of MSTE showed that LIPUS significantly increased the FCMC between the motor cortex and the tail muscles. The FCMC of LFP-EMG after LIPUS was significantly higher than that of EMG-LFP. Through the change of NTB, the FCMC between the motor cortex and tail muscle also changed. As FCMC is a promising tool for motor evaluation, it is proved that LIPUS can affect the motor function of mice. The multi-scale transfer entropy analysis method introduces a time scale, which provides a reference for more factors to evaluate the effect of LIPUS on neuromodulatory.

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